

TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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This supplement contains the inherited NBO proofs, detailed records for the corrected complete-service experiment, the path-accounting correction and the source-preservation map. The preceding complete article and broader development remain unchanged. The current numerical claims concern the original expected-cost investment model and the declared two-control extension.

1. PROOFS OF THE INHERITED NBO RESULTS

These proofs retain the original statements and arguments. Their placement in the supplement changes the reading order, not their hypotheses.

1.1. *Proof of the result 1*

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t . The terminal width is unchanged. Q.E.D.

1.2. *Proof of the result 2*

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t . At a nearest node its cone exceeds the infimum by

at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \text{ReLU}(z) + \text{ReLU}(-z)$ and $\min(u, v) = u - \text{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. Q.E.D.

1.3. Proof of the result 3

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x ; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. Q.E.D.

1.4. Proof of the result 4

PROOF: The state cost has gradient infinity norm at most $8/d$, Hessian norm at most $22/d$, and diagonal second derivatives at most $5/d + 16/d^2$. For the terminal cost the corresponding bounds are $10/d$, $26/d$, and $9/d + 16/d^2$. They follow by differentiating the quadratic terms and the continuously differentiable squared shortage. The bound on the cycle Laplacian is four. The shortage Hessian, where it exists, is $16\mathbf{1}\mathbf{1}^\top / d^2$; the same second-difference bound holds across its kink.

For feasible (x, a) and (y, b) , put $w = x - y$. The difference between the transition at their convex combination and the corresponding combination of transitions is $\theta(1 - \theta)w_j w_{j+1} / 16$ in coordinate j . Convexity and the G_{t+1} Lipschitz bound

therefore give a convexity defect of at most

$$\frac{\beta G_{t+1}}{16} \theta (1 - \theta) \sum_j |w_j w_{j+1}| \leq \frac{\beta G_{t+1}}{16} \theta (1 - \theta) \|w\|_2^2.$$

The state cost supplies curvature $4/d$, while the two-control action cost has Hessian eigenvalues at least $7/4$. Thus the state curvature of Q_t^* is at least $4/d - \beta G_{t+1}/8$, which is at least $107/(128d)$ when $G_{t+1} \leq 27/d$. Partial minimization over the affine capacity graph preserves the state convexity inequality. The minimizing action is unique, including at a binding capacity constraint.

To propagate Lipschitz continuity, write $a = c(x)u$, where u lies in the fixed unit simplex. Transfer the same u from x to y . Each action-cost partial derivative has absolute value at most $13/16$, and $\sum_j u_j \leq 1$. The action-cost transfer contributes at most $13\|x - y\|_1/(128d)$. The transition Jacobian at a fixed action has column sums at most $11/16$; the capacity transfer adds at most $3/64$. Comparing a minimizing normalized action in both directions gives the stated recursion for G_t . Its right side maps $27/d$ below itself.

For a fixed u , the transition as a function of x has Jacobian norm at most $11/16 + 1/16 = 3/4$. Its second-order composition term is bounded by $G_{t+1}/8$. The action-cost Hessian has norm at most $21/4$, and the capacity gradient has squared norm $1/(64d)$. These facts give the recursion for M_t . The infimum over u preserves this common semiconcavity bound: after subtracting the same quadratic, each fixed- u function is concave, and an infimum of concave functions is concave. The right side maps $54/d$ below itself. Convexity together with semiconcavity yields the claimed differentiability and gradient regularity; this conclusion does not require a differentiable optimizing-action map.

Along one state coordinate, the fixed- u transition has zero second derivative. The squared norm of its first derivative is at most $85/256 + 23/(256d)$: the two original nonzero entries are bounded by $9/16$ and $1/8$, and the capacity-induced entries are bounded by $1/(16d)$. Combining this with the diagonal stage-cost bound gives D_t . Finally, positive interpolation along one coordinate has error between zero and $D_t h^2/8$. Applying the coordinate interpolations successively and adding the bounds proves (10). Q.E.D.

1.5. Proof of the result 5

PROOF: Joint convexity, including the feasible graph, gives

$$Q_t^*(x, \bar{\pi}_t(x)) \leq \sum_v w_v(x) Q_t^*(v, a_t(v)) \leq I_n q_t^{\text{sel},+}(x).$$

The lower interpolation bound gives $V_t^*(x) \geq I_n l_t(x) - \kappa_t$. Subtraction proves the uniform optimal-advantage bound e_t . The partial action derivatives of Q_t^* are bounded by Λ_t , so feasible implementation error adds at most $\Lambda_t b_t$. Adding the future policy loss and proceeding backwards proves the recursion. No convexity of $J^{\hat{\pi}}$ is used. Q.E.D.

1.6. Proof of the result 6

PROOF: Apply concavity to $f(a) - Ha^2/2$ and substitute the two lower endpoints. The displayed quadratic has derivative $D - K + 2Ks$. Minimizing it on the unit interval gives the three cases. Its coefficient of K is $-s(1 - s)$, which proves the last assertion. Q.E.D.

1.7. Proof of the result 7

PROOF: At $r = 1$, analytic endpoint enclosures and the minorant lemma give the assertion. Suppose it holds for $r - 1$. Each child interval encloses the true optimal continuation. If the numerical next state is an interval, expanding the queried center value by G_{r-1} times the maximum ℓ^1 displacement encloses all its possible values. Conditional-mean lower bounds and the semiconcavity remainder then give valid endpoint brackets for the true Q_r . The minorants cover the entire action interval. A downward capacity endpoint can omit only an interval of length equal to the certified capacity-rounding difference; subtracting Λ_r times that difference accounts for its possible improvement. The minimum upper endpoint is the cost of a feasible first action followed by the optimal future. This proves (i) by induction. Retaining earlier valid lower bounds preserves every conclusion.

For termination, an interval of length h has minorant at least its smaller lower endpoint minus $H_r h^2/8$. Its smaller upper endpoint is an available global upper bound. With the endpoint-width guard, an interval of length at most $h_\delta = \sqrt{3\delta/H_r}$

cannot be responsible for a gap exceeding δ . Every refined parent is therefore longer than h_δ . Its children are longer than $h_\delta/5$. Apart from the at most two initial fragments made by a prediction, disjoint terminal fragments consequently have total number at most $5c(x)/h_\delta$, up to endpoint counting. Equation (15) is a conservative point budget. Bounded action-quantum rounding of a proposed quarter-to-three-quarter split preserves the one-fifth property whenever the quantum is smaller than $h_\delta/20$. Otherwise precision must be increased; a fixed-precision run is not allowed to ignore this requirement.

For (iii), let $\bar{x}$ be the acquired state. Its downward displacement is at most $d2^{-b}$ in ℓ^1 , and $c(\bar{x}) \leq c(x)$ preserves feasibility. At a fixed action the stage cost is $8/d$ -Lipschitz and the transition has column sums at most $11/16$. Thus replacing $\bar{x}$ by x changes Q_r by at most $(8 + 11\beta dG_{r-1}/16)2^{-b}$. The corresponding value change is at most $dG_r 2^{-b}$. Add these two allowances to (i). Substituting the policy's own future yields the usual one-step loss recursion; backward induction gives (16). No policy-value oracle and no population claim about fitting are used. *Q.E.D.*

1.8. Proof of the result 8

PROOF: A routed action is merely another feasible partition point. The endpoint and minorant induction in Theorem 7 therefore applies without change. After each evaluation retain the smallest upper endpoint and the maximum of the old and newly valid global lower endpoints. Their difference cannot increase. The stopping test uses this difference, not a fitted value or a prediction error.

Put $h_\delta = \sqrt{3\delta/H_r}$. A partition interval of length at most h_δ has minimum minorant at least its smaller lower endpoint minus $3\delta/8$. The associated upper endpoint, with the stipulated width and capacity allowance, is within δ of that minorant. Such an interval cannot be responsible for an unresolved global gap. Every split parent therefore has length greater than h_δ . The one-fifth condition makes each of its children longer than $h_\delta/5$. Starting with one interval, disjoint terminal fragments consequently number at most $1 + 5c(x)/h_\delta$. Counting endpoints and allowing for rounding of this numerical bound gives the displayed conservative budget. Floating

failure is handled by the separately charged exact-rational procedure, not by this real-arithmetic count.

The first assertion about predictor evaluation follows from the order of the stopping test and the routing rule. For the last assertion, use backward induction on the remaining horizon, with the common analytic terminal kernel as base. At each nonterminal query, declined predictions leave the same conventional split, lower bound, witness and tie rule. Induction over partition refinements gives the same next query and returned endpoints; the induction over dates then gives the same implemented policy. Model-evaluation work can still differ, so transcript identity is not a zero-cost fitting claim. *Q.E.D.*

1.9. Proof of Proposition 9

PROOF: Every retained lower endpoint is at least its true endpoint value minus w , and hence at least $f(a^*) - w$. On an interval of length at most h , the parabolic minorant is therefore at least $f(a^*) - w - Hh^2/8$. Subtracting the omitted-capacity allowance gives $L \geq f(a^*) - w - Hh^2/8 - \Lambda\Delta$; retaining an older, larger valid lower bound only improves this inequality. The upper endpoint at $\hat{a}$ is at most $f(\hat{a}) + w$. Semiconcavity and the one-sided first-order condition give $f(\hat{a}) - f(a^*) \leq \lambda e + He^2/2$. Subtract the lower bound to prove (18). At an interior optimum the derivative is zero. The example $f(a) = a$ on $[0, c]$ shows that the boundary linear term is needed. Finally, retaining bounds makes $U' \leq U$ and $L' \geq L$; expansion proves (19). *Q.E.D.*

1.10. Proof of the result 14

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i , and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of Maurer and Pontil (2009) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every

$\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

1.11. Proof of the result 15

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (34). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

1.12. Proof of the result 16

PROOF: Subtract the common J_{b_0} . The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ . A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\hat{b}(\lambda_2)} - s_{\hat{b}(\lambda_1)}) \leq 0$. Since s_b increases strictly with b , the precision comparative static follows. *Q.E.D.*

2. NUMERICAL ACCOUNTING AND INDEPENDENT CHECKS

The source-frozen experiment preceding this revision used a cumulative path-cost interval initialized by a point constructor. That constructor can share its lower and upper arrays. The subsequent statements assigning a new lower endpoint and then a new upper endpoint therefore left both running endpoints equal to the latter. This is an enclosure error even when its magnitude is small. It cannot be justified by the statistical interval being much wider.

For the first designed zero-state path in the two-state, two-date adaptive comparison, the original archived cost lower endpoint is approximately 2.6744468902315184. An independently evaluated exact rational midpoint trajectory has cost approximately 2.6744468902315166, below that endpoint. The complete rational value, raw file hash, action sequence and original endpoint are recorded in the erratum audit. The midpoint is used here only to refute the claimed numerical enclosure; it is not used as an exact expectation under the continuous law.

The repaired accumulator uses two independently allocated arrays. It retains outward arithmetic at every addition and multiplication and re-encloses the entire continuous bin. The original source, observations and process receipts remain in the R66 directory. The corrected source is committed and frozen before the full catalogue is rerun into a new directory. The two executions share their declared random indices, so only one set of inference rows is counted. An exact regression reproduces the aliasing risk and checks the repaired path against independent rational primitives. The publication audit checks all path enclosures and distinguishes this check from re-integrating a complete continuation tree.

2.1. *Reproduction boundary*

Every declared receipt is retained, including failed or timed-out services. Source and dependency hashes are checked against the pre-execution freeze. Model hashes, stored acquisition states, action lattices, true-state capacity, local gaps and complete cost endpoints are checked for every archived path. All ninety distinct scalar comparison services are re-queried from their saved models; the second clock repetition must reproduce the same model and transcript. This re-query uses the original numerical kernel and is a reproduction test, not an independent proof of that kernel. The large vector continuation services and the inference and reuse endpoint queries are not independently re-integrated in the publication replay.

Every path archive is recomputed with the corrected interval primitives and its recorded actual actions. Two fixed midpoint trajectories per archive or inference batch are additionally evaluated using separately written rational stage costs and transitions. Midpoint checks supplement, but do not replace, whole-bin interval verification. The tests of primitive identities, triangular lower bounds, terminal moments, simplex projection, source contracts and exact readout optimization are separate from scientific observations. They do not increase the number of training seeds or policy-cost samples.

3. COMPLETE PROGRESS, SEEDS AND PAIRED COSTS

TABLE I: Where verification work makes progress

d/T	Selector	Queries	Routed	Zero	Upper	Lower	Both	Screen
2/2	Adaptive	106	0	0	0	61	45	3
2/2	Bisection	124	0	0	0	72	52	0
2/2	Upper chord	114	0	0	0	70	44	1
2/2	ReLU	814	231	0	0	540	274	0
2/2	Quadratic	828	249	0	0	541	287	0
2/2	Linear	830	264	0	0	530	300	0
2/2	Spline	822	227	0	0	550	272	0
2/2	Nearest	866	230	0	0	532	334	0
2/2	Derivative	116	28	0	0	51	65	0
2/2	Midpoint	117	46	0	0	63	54	0
8/3	Adaptive	745	0	0	0	580	165	6
8/3	Bisection	1010	0	0	0	826	184	18
8/3	Upper chord	789	0	0	0	609	180	0
8/3	ReLU	6016	871	0	0	4954	1062	16
8/3	Quadratic	5981	859	0	0	5017	964	19
8/3	Linear	5972	970	0	0	4719	1253	9
8/3	Spline	6056	817	0	0	5129	927	25
8/3	Nearest	6126	934	0	0	4778	1348	18
8/3	Derivative	792	141	0	0	580	212	0
8/3	Midpoint	911	488	0	0	725	186	0

Only one copy of each distinct transcript is counted. Queries denotes new refinements, not initial boundary queries. Zero counts routed queries that improve neither global endpoint; Upper, Lower and Both classify all refinements. Screen counts returns without a new endpoint query. Numeric upper and lower increments and every zero-progress row remain in the raw records. The upper and lower improvements are recorded separately at every recursive query. The global lower endpoint retains all previously valid bounds. The displayed counts use one copy of each distinct transcript; the raw archives retain both clock repetitions. A screen return has an outward upper witness for the actual rounded action and a valid global

TABLE II

ALL PAIRED EXPECTED-COST DIFFERENCES

d/T	Difference	Simultaneous interval
2/2	Adaptive – Bisection	[-0.02642, 0.02643]
2/2	Adaptive – Quadratic	[-0.02638, 0.02646]
2/2	Adaptive – ReLU	[-0.02638, 0.02646]
2/2	Bisection – Quadratic	[-0.02639, 0.02646]
2/2	Bisection – ReLU	[-0.02638, 0.02646]
2/2	ReLU – Quadratic	[-0.02642, 0.02642]
8/3	Adaptive – Bisection	[-0.03302, 0.03303]
8/3	Adaptive – Quadratic	[-0.03301, 0.03303]
8/3	Adaptive – ReLU	[-0.03301, 0.03304]
8/3	Bisection – Quadratic	[-0.03301, 0.03303]
8/3	Bisection – ReLU	[-0.03301, 0.03303]
8/3	ReLU – Quadratic	[-0.03302, 0.03302]

Every interval is formed from paired path endpoints, not by subtracting the printed marginal confidence intervals. All six contrasts per task are retained. The sample stream is common within a task; method arms are not additional independent rows.

lower bound. The audit verifies its score and the saved-model re-query verifies the child tolerance at which it returned.

TABLE III: Every fitting seed and its complete process cost

d/T	Selector	Seed	Queries	Full (s)	Warnings
2/2	ReLU	6601	1209.0	0.766	0
2/2	ReLU	6602	1209.0	0.766	0
2/2	ReLU	6603	1204.0	0.766	0
2/2	ReLU	6604	1209.0	1.017	2
2/2	ReLU	6605	1204.0	0.791	0
2/2	ReLU	6606	1219.0	0.766	2
2/2	ReLU	6607	1214.0	0.791	0
2/2	ReLU	6608	1234.0	0.766	2
2/2	Quadratic	6601	1209.0	0.365	0

	d/T	Selector	Seed	Queries	Full (s)	Warnings	
1							1
2	2/2	Quadratic	6602	1224.0	0.365	0	2
3	2/2	Quadratic	6603	1229.0	0.365	0	3
4	2/2	Quadratic	6604	1229.0	0.365	0	4
5	2/2	Quadratic	6605	1214.0	0.365	0	5
6	2/2	Quadratic	6606	1214.0	0.365	0	6
7	2/2	Quadratic	6607	1214.0	0.365	0	7
8	2/2	Quadratic	6608	1239.0	0.365	0	8
9	2/2	Linear	6601	1214.0	0.365	0	9
10	2/2	Linear	6602	1214.0	0.365	0	10
11	2/2	Linear	6603	1224.0	0.365	0	11
12	2/2	Linear	6604	1224.0	0.365	0	12
13	2/2	Linear	6605	1224.0	0.365	0	13
14	2/2	Linear	6606	1219.0	0.365	0	14
15	2/2	Linear	6607	1224.0	0.365	0	15
16	2/2	Linear	6608	1239.0	0.365	0	16
17	2/2	Spline	6601	1234.0	0.365	0	17
18	2/2	Spline	6602	1214.0	0.365	0	18
19	2/2	Spline	6603	1209.0	0.365	0	19
20	2/2	Spline	6604	1209.0	0.365	0	20
21	2/2	Spline	6605	1214.0	0.365	0	21
22	2/2	Spline	6606	1219.0	0.365	0	22
23	2/2	Spline	6607	1214.0	0.365	0	23
24	2/2	Spline	6608	1229.0	0.365	0	24
25	2/2	Nearest	6601	1249.0	0.365	0	25
26	2/2	Nearest	6602	1239.0	0.365	0	26
27	2/2	Nearest	6603	1239.0	0.365	0	27
28	2/2	Nearest	6604	1244.0	0.365	0	28
29	2/2	Nearest	6605	1249.0	0.365	0	29
30	2/2	Nearest	6606	1244.0	0.365	0	30
31	2/2	Nearest	6607	1239.0	0.365	0	31
32	2/2	Nearest	6608	1259.0	0.365	0	32
	8/3	ReLU	6601	14496.0	2.570	4	
	8/3	ReLU	6602	14343.0	2.495	4	

	d/T	Selector	Seed	Queries	Full (s)	Warnings	
1							1
2	8/3	ReLU	6603	14614.0	2.645	4	2
3	8/3	ReLU	6604	14768.0	2.670	4	3
4	8/3	ReLU	6605	14660.0	2.645	4	4
5	8/3	ReLU	6606	14433.0	2.520	4	5
6	8/3	ReLU	6607	14524.0	2.620	4	6
7	8/3	ReLU	6608	15058.0	2.695	4	7
8	8/3	Quadratic	6601	14786.0	2.219	0	8
9	8/3	Quadratic	6602	14370.0	2.069	0	9
10	8/3	Quadratic	6603	14578.0	2.119	0	10
11	8/3	Quadratic	6604	14343.0	2.094	0	11
12	8/3	Quadratic	6605	14895.0	2.219	0	12
13	8/3	Quadratic	6606	14633.0	2.219	0	13
14	8/3	Quadratic	6607	14560.0	2.220	0	14
15	8/3	Quadratic	6608	14352.0	2.069	0	15
16	8/3	Linear	6601	14878.0	2.169	0	16
17	8/3	Linear	6602	14899.0	2.219	0	17
18	8/3	Linear	6603	14624.0	2.194	0	18
19	8/3	Linear	6604	14407.0	2.119	0	19
20	8/3	Linear	6605	14407.0	2.094	0	20
21	8/3	Linear	6606	14823.0	2.194	0	21
22	8/3	Linear	6607	14652.0	2.119	0	22
23	8/3	Linear	6608	14326.0	2.094	0	23
24	8/3	Spline	6601	14469.0	2.069	0	24
25	8/3	Spline	6602	14446.0	2.069	0	25
26	8/3	Spline	6603	14686.0	2.194	0	26
27	8/3	Spline	6604	14817.0	2.194	0	27
28	8/3	Spline	6605	14636.0	2.219	0	28
29	8/3	Spline	6606	14650.0	2.219	0	29
30	8/3	Spline	6607	14677.0	2.219	0	30
31	8/3	Spline	6608	14695.0	2.219	0	31
32	8/3	Nearest	6601	15108.0	2.219	0	32
	8/3	Nearest	6602	14669.0	2.245	0	
	8/3	Nearest	6603	14715.0	2.169	0	

d/T	Selector	Seed	Queries	Full (s)	Warnings
8/3	Nearest	6604	14904.0	2.219	0
8/3	Nearest	6605	14949.0	2.245	0
8/3	Nearest	6606	14605.0	2.169	0
8/3	Nearest	6607	15012.0	2.169	0
8/3	Nearest	6608	14424.0	2.044	0

Each row contains the median of two clock repetitions of that declared fitting seed. The repetitions reproduce a model and transcript and are not two independent fitting initializations. All warnings, including lack of optimizer convergence, are retained. For a matching seed, all five fitted selectors use the same ninety-six label states. The states are regenerated from the fixed training seed, with four named boundary and interior rows. The ReLU configuration is width twenty-four and the declared L-BFGS stopping rule; it is not replaced by the new convex readout optimizer after seeing the results. Least-squares, spline and nearest-action models retain their declared regularization or sample rule. Neither a warning nor a failed candidate is removed from its process cost. A finite collection of initializations is not a population training-convergence theorem.

4. CACHED REUSE AND STATISTICAL INTERPRETATION

TABLE IV: Measured reuse crossings without extrapolation

d/T	Selector	Seed	Rep.	First	Persistent	Full win
2/2	Bisection	0	0	–	–	no
2/2	Bisection	0	1	–	–	no
2/2	ReLU	6603	0	–	–	no
2/2	ReLU	6603	1	–	–	no
2/2	ReLU	6607	0	–	–	no
2/2	ReLU	6607	1	–	–	no
2/2	Quadratic	6603	0	–	–	yes
2/2	Quadratic	6603	1	24	24	yes

<i>d/T</i>	Selector	Seed	Rep.	First	Persistent	Full win
2/2	Quadratic	6607	0	–	–	yes
2/2	Quadratic	6607	1	–	–	yes
8/3	Bisection	0	0	–	–	no
8/3	Bisection	0	1	–	–	no
8/3	ReLU	6603	0	–	–	no
8/3	ReLU	6603	1	–	–	no
8/3	ReLU	6607	0	–	–	no
8/3	ReLU	6607	1	–	–	no
8/3	Quadratic	6603	0	–	–	no
8/3	Quadratic	6603	1	–	–	no
8/3	Quadratic	6607	0	–	–	no
8/3	Quadratic	6607	1	–	–	no

First and Persistent are block counts within the fixed twenty-four-block window, for cumulative clocks through durable block records. Persistent means no reversal within the observed remainder, not beyond it. Full win uses the separately recorded launch-through-exit process time. A dash means no crossing observed. The stale-identity block changes the loaded acquisition-contract identifier, not the stored file. The failed identity check selects the conventional adaptive verifier and is timed together with its workload. A model-specific source digest and acquisition precision are part of the identity. The experiment therefore tests an actual rejected reuse, not a hypothetical cost inserted into an amortization formula.

The inference rows are separate from the named workload diagnostics. Their initial bins cover the full continuous state cube with the declared uniform law. Within-bin uncertainty is propagated through discontinuous acquired actions by checking whether the entire state enclosure has one acquisition code. Ambiguous paths receive full remaining support. Mean and variance enclosures are computed outward, using a conservative square-root endpoint and the exact logarithm allowance. Centering the sum of squares at an outward mean midpoint can only increase the exact sum relative to centering at the exact sample mean; it therefore

gives a valid upper variance allowance. The same finite-family bound covers selecting a policy by its upper cost endpoint after observing the simultaneous intervals.

Expected nonworsening or a uniform Bellman gap does not imply pathwise dominance under a common shock. The paired path support retains both signs. The complete-service purchase formula in the article is a partial-identification statement conditional on the declared prices, not an empirical estimate of those prices.

5. PRESERVED CONTENT AND COMPUTATIONAL SCOPE

The original title, own-future continuation, feasible witness construction, original-optimum benchmark, acquisition results and policy-inference statements remain in the active manuscript. Their proofs above are relocated without changing their arguments. The preceding article and technical supplement remain byte-for-byte in the R65 directory and are included as companion PDFs. Its complete R62 development editions remain available with their full original assumptions. The content map records every prior source hash and label location; older adverse comparisons are not deleted.

The current scalar experiments and the two-control extension use linear conditional expectation. The wider diffusion, recursive-utility, temporal-self and game results in the retained development are not silently covered by this query theorem. Likewise, a fixed-feature convex ReLU readout is not unrestricted nonconvex hidden-layer training. Its achieved objective and dual gap can be certified exactly, while a distributional work assertion additionally requires the independent context law stated in Theorem 11. The raw recursive transcript alone does not establish that law.

REFERENCES

MAURER, ANDREAS AND MASSIMILIANO PONTIL (2009): “Empirical Bernstein Bounds and Sample Variance Penalization,” in *Proceedings of the 22nd Annual Conference on Learning Theory*, arXiv:0907.3740. [6]